

Week 3, Lecture 2 WOD: Answer Key

A Probability Puzzle

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. The guesses (1a, Part 3) have no right answer at the time they're made. Part 3 is worked out in full at the end, for whenever you reveal it in class. All numbers are computed in R when this file is knitted.

Part 1: Sevens vs. doubles

1a.

Before doing any counting: which do you think is more likely, rolling a **sum of 7**, or rolling **doubles** (both dice show the same number)? Circle one: sum of 7 / doubles / equally likely

Answer. Any guess is fine. The correct answer (1b-1d) is **equally likely**.

Why. The point is to get a guess on record before counting, so students can see where their intuition was off.

Watch for. Students who skip ahead and count first. Ask them to circle their gut answer anyway.

1b.

How many equally-likely outcomes are there in the full sample space Ω (all ordered pairs of red/blue results)? $|\Omega| = \underline{\hspace{2cm}}$

Answer. 36.

Why. 6 outcomes for the red die times 6 for the blue die. The dice are distinguishable, so (2, 5) and (5, 2) are different outcomes.

Watch for. Answering 21, which counts unordered pairs. Those 21 outcomes aren't equally likely: (2, 5) can happen two ways, but (3, 3) only one.

1c.

How many of those outcomes sum to 7? List them if it helps. $P(\text{sum} = 7) = \underline{\hspace{2cm}}$

Answer. 6 outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1). $P(\text{sum} = 7) = 6/36 = 1/6$.

Why. Every red value from 1 to 6 has exactly one blue value that makes 7.

Watch for. Listing only 3 outcomes (treating (1,6) and (6,1) as the same) and getting 3/36.

1d.

How many outcomes are doubles? $P(\text{doubles}) = \underline{\hspace{2cm}}$

Answer. 6 outcomes: (1,1) through (6,6). $P(\text{doubles}) = 6/36 = 1/6$.

Why. Each red value has exactly one matching blue value.

Watch for. Students who got 3/36 in 1c but 6/36 here are counting the two events inconsistently.

1e.

Was your guess in 1a correct? If not, what made doubles or sevens feel more/less likely than it actually is?

Answer (discussion). They're tied, at 6 of 36 each.

Why. Common reasons for a wrong guess: 7 is famous as the most common sum (true), so it feels like it should beat everything; doubles feel special, so they feel rare. Worth pointing out: the tie happens only because 7 is the most common sum. Any other sum, such as 8 (5 ways), loses to doubles.

```
sums <- outer(1:6, 1:6, "+")
table(sums)
```

```
## sums
##  2  3  4  5  6  7  8  9 10 11 12
##  1  2  3  4  5  6  5  4  3  2  1
```

Watch for. “I guessed right” with no reason. Ask what they would have guessed for a sum of 8.

Part 2: How many tickets?

A small regional lottery asks you to pick 5 distinct numbers from 1 to 30 (order doesn't matter). How many different tickets are possible? Leave your answer as an expression (use factorials!).

Answer.

$$\binom{30}{5} = \frac{30!}{5!25!} = \frac{30 \times 29 \times 28 \times 27 \times 26}{5!} = 142,506$$

(The integer isn't required.)

Why. There are $30 \times 29 \times 28 \times 27 \times 26 = 30!/25!$ ways to write down 5 distinct numbers in order. Each ticket appears $5! = 120$ times in that list (once per ordering), so divide by $5!$.

Watch for.

- Stopping at $30!/25!$ (the ordered count), which is 120 times too big. This is what the hint's “how do you account for order” is getting at.
- Writing 30^5 , which allows repeated numbers.

Part 3: A teaser (guess only)

Suppose there are 25 people in this room. Do you think it's more likely than not (better than a 50% chance) that **at least two of them share a birthday**? Circle your guess: yes / no

Answer. Any guess is fine at the time. The correct answer is **yes**: about 56.9%.

Why. The full calculation is below. The short version: what matters is the number of *pairs* of people (300), not the number of people compared with 365 days.

Watch for. Most students guess “no,” comparing 25 with 365. That's the comparison the reveal should correct.

Full worked solution, for the reveal

Assume 365 equally likely birthdays (ignore Feb. 29) and independence between people.

Use the complement. “At least one match” can happen in many ways (one pair, two pairs, three people on one day, ...). “No matches” happens in only one way, so compute that and subtract:

$$P(\text{at least one match}) = 1 - P(\text{no matches})$$

Build “no matches” one person at a time. Given that everyone so far has a different birthday, person k must avoid the $k - 1$ days already taken:

- Person 1: 365/365
- Person 2: 364/365
- Person 3: 363/365
- ...
- Person 25: 341/365

Multiply these conditional probabilities (the same chaining used for the detection-classifier calculation in Lab 3, Act 1):

$$P(\text{no matches}) = \prod_{k=1}^{25} \frac{366 - k}{365} = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{341}{365}$$

```
n <- 25
p_no_match <- prod((365 - 0:(n - 1)) / 365)
c(p_no_match = p_no_match, p_match = 1 - p_no_match)

## p_no_match    p_match
## 0.4313003    0.5686997
```

$P(\text{no matches}) \approx 0.431$, so $P(\text{at least one match}) \approx 0.569$.

Why it feels wrong. Any pair of people could be the match, and 25 people make many pairs:

```
choose(25, 2)
```

```
## [1] 300
```

Each pair has only a 1/365 chance of sharing a birthday, but 300 chances add up quickly.

Optional aside: computing in log space. The same answer can be written with factorials, $P(\text{no matches}) = \frac{365!}{340! 365^{25}}$, but 365! is far too large for a computer to store. Taking logs avoids that (the same trick as the log-likelihood later in the course):

```
exp(lfactorial(365) - lfactorial(365 - n) - n * log(365))
```

```
## [1] 0.4313003
```

For context during the reveal:

```
ns <- c(10, 15, 20, 23, 25, 30, 40, 50)
data.frame(n = ns,
           P_match = round(1 - sapply(ns, function(m) prod((365 - 0:(m - 1)) / 365)), 3))
```

```
##      n P_match
## 1 10  0.117
## 2 15  0.253
## 3 20  0.411
## 4 23  0.507
## 5 25  0.569
## 6 30  0.706
## 7 40  0.891
## 8 50  0.970
```

The commonly cited tipping point is $n = 23$. The WOD asks about 25 so students can't just recall the classic number.